

'on' position. All the 5mm white LEDs (LED1 through LED22) glow. To get a higher light intensity, you can use a reflector. Similarly, if you want to use the lamp at night, connect a 12V battery to the anode of diode D1.

For using the variable power supply, flip switch S2 to 'on' position. Connect a multimeter across CON1 terminals and set the required voltage (say, 9V) using potmeter VR1. The 9V output will be available across CON1 terminals. As mentioned earlier, the range of the power supply is 1.5V to 15V.

For charging a Li-ion battery (used in mobile phones), flip switch S3 to 'on' position and use connector CON2. Zener diode ZD1 provides 4.7V to charge the Li-ion battery. Using this circuit, you can charge a 3.6V Li-ion cell very easily. Resistor R13 limits the charging current.

Assemble the circuit on a general-purpose PCB and enclose in a small box. Mount RCA socket on the front panel of the box. Use RCA plug with cable for connecting the battery and LEDs. Also mount all the switches on the front panel of the cabinet. ●

Smart Bicycle Light



T.K. HAREENDRAN

With this circuit built, boxed up and installed in your bicycle, you can look forward to safely riding your cool bicycle at night. The circuit ensures that the bicycle lights remain 'on' for a little while after the bicycle has come to a standstill.

The circuit is 6V, 3W dynamo compatible and its working is very simple and straightforward. When the bicycle dynamo is idle, it connects the light source to the reservoir battery pack until the dynamo starts generating voltage again. The circuit also prevents the light source against high power when the dynamo is running, and uses the available surplus power to recharge the battery pack.

As the diagram shows, the standard bicycle headlight bulb is replaced with a compact 1W generic white LED module. Likewise, the tail lamp is replaced with a 10mm red LED. The battery pack is a combination of five 1.2V, 600mAh Ni-Cd cells, connected in series to produce an output of 6V DC.

When you ride the bicycle, the dynamo touching the wheel spins around and diodes D1, D2, D3 and D4 form a bridge via which buffer capacitor C1 charges. When the dynamo voltage is sufficiently high, the low-current 5V relay energises and the DC supply available across C1 is routed to the head lamp (LED1) driver circuit (and to the tail lamp LED) through power switch S1. At the same time, the Ni-Cd battery pack charges through components D5 and R1.

When the bicycle is not driven and the dynamo is standstill, the relay is no longer energised. Therefore the bicycle lights (LED1 and LED2) are powered by the Ni-Cd battery pack. Diode D5 prevents the Ni-Cd battery from slowly discharging via the circuit when the bicycle is not used (switch S1

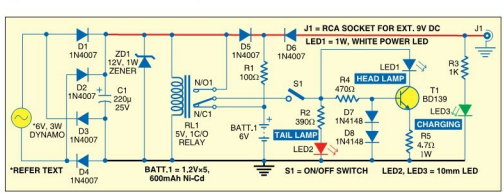


Fig. 1: Circuit for smart bicycle light